OFFICIAL DOCUMENTATION

Clearpane — User Documentation

A calm, minimal task manager built for deliberate personal productivity

Stack: Next.js 16 • React 19 • Supabase • Tailwind CSS

Security: Google OAuth 2.0 • PostgreSQL Row-Level Security

Specification: WCAG 2.1 AA Accessible

1. How to access and use the application

Quick Start Overview

Clearpane can be accessed from any desktop, tablet, or mobile browser by visiting `https://kovai-io.vercel.app`.

The Dashboard Bento Grid

Upon authenticating, users land on the main workspace at `/dashboard`, structured as a clean visual bento grid:

- **Greeting & Quick Add:** Welcomes you based on your browser's local time (morning, afternoon, or evening). Type any task idea into the input and press `Enter` to launch the task creation dialog pre-filled with your idea.
- **Completion Ring:** An interactive, animated circular gauge reflecting your overall completion percentage (e.g., "3 of 5 tasks complete"). Fully disabled under reduced-motion settings.
- **Status Distribution Chips:** Clickable statistics chips displaying counts for Planned, In Progress, and Complete tasks. Clicking any chip navigates to `/tasks` with that filter pre-selected.
- **Progress Breakdown:** A proportional multi-colored bar chart visualizing status distribution across your workspace.
- **In Focus Now:** Highlights tasks currently in progress, allowing one-click completion.
- **Recent Activity:** Chronological audit log showing status changes and timestamps.

Managing Tasks

- **Creating a Task:** Click **New task** or press the keyboard shortcut `N` anywhere in the app. Enter a required Title (1–120 characters) and optional Notes (up to 1,000 characters). Select an initial status (defaults to Planned) and click **Create Task**.
- **Changing Status:** On any task card, click the 3-option segmented control (*Planned*, *In Progress*, *Complete*). State updates optimistically in under 16ms with automatic rollback if a network error occurs. Keyboard users can navigate status options using arrow keys.
- **Filtering & Searching:** On `/tasks`, switch between filter tabs (All, Planned, In Progress, Complete) with live counts. The search input filters by title in real-time with a 200ms debounce.

- **Editing & Deleting:** Click the card options menu (`...`) to edit details or delete. Deletion triggers an instant 6-second undo window with a toast notification to prevent accidental loss.

2. Login instructions

Authentication Steps

1. Navigate to `/login` (or any protected page, which automatically redirects to login).
2. Click **Continue with Google**.
3. Select your Google account on the Google OAuth authorization screen.
4. Review the requested permissions:
   - **Email address:** Used exclusively to identify your private account.
   - **Basic Profile (Name & Picture):** Used strictly to personalize your dashboard greeting and user avatar.
   - Clearpane **never** requests access to Google Drive, Gmail, Contacts, or Calendar.

> **Handling the "Google hasn't verified this app" screen:**
> Because Clearpane is an educational evaluation project, the Google OAuth Client may be in Testing mode. If Google displays an "Unverified App" warning:
>
> 1. Click **Advanced** in the bottom-left of the warning prompt.
> 2. Click **Go to Clearpane (unsafe)** to proceed with login.
> 3. Confirm that your email address is listed as a Test User in the Google Cloud project.

Troubleshooting Login Issues

- **Sign-in Cancelled or Failed:** If you cancel or Google encounters an error, you will be redirected to `/login?error=auth_failed` with a friendly alert banner. Simply retry the sign-in.
- **Third-Party Cookie Blockers:** Ensure your browser allows cookies from your Supabase domain, or temporarily disable strict tracker blockers that intercept OAuth PKCE exchanges.

3. Important assumptions made

To ensure a calm, deliberate, and high-quality user experience, the following architectural assumptions were established:

- **Strict Account Privacy:** Tasks are private to each Google account. No user can view, edit, or search another user's tasks. Data isolation is enforced by database-level PostgreSQL Row-Level Security (RLS).
- **Field Boundaries & Validation:**
  - **Title:** Required, trimmed of leading/trailing whitespace, between 1 and 120 characters. Whitespace-only titles are rejected.
  - **Notes:** Optional, up to 1,000 characters. Empty or whitespace-only notes are automatically cleaned and stored as `NULL`.
- **Default Status:** All newly created tasks default to **Planned** unless explicitly set otherwise during creation.
- **Any-to-Any Status Transitions:** Work is flexible. Tasks can transition directly between any state (e.g. from Planned straight to Complete, or from Complete back to In Progress).
- **Chronological Ordering:** Tasks are presented newest first, ordered by `created_at DESC`.
- **Edit and Delete Usability:** While the core assessment specified create, view, and update status, edit and soft-undo delete were added to deliver a complete, realistic user experience.
- **Local Time Display:** Timestamps are stored in UTC and rendered in your browser's local time zone to eliminate hydration mismatches and time zone confusion.
- **Language:** The interface and validation messages are presented in English.

4. Known limitations

- **No Multi-User Teams or Sharing:** Clearpane is intentionally built as an individual productivity workspace; shared workspaces, task assignments, and public lists are not included.
- **No Due Dates or Notifications:** Clearpane deliberately avoids deadline anxiety; due dates, reminders, and calendar synchronization are excluded.
- **No In-App Account Deletion:** Users cannot delete their Google Auth record from within the application interface (handled via the Supabase Auth administration dashboard).
- **Concurrent Multi-Tab Editing (Last Write Wins):** If Clearpane is open across multiple browser tabs, updates apply on a "last write wins" basis. Background tab refocus refreshes data automatically (throttled to once per 10s).
- **Tab Closure During Undo:** The 6-second undo delete relies on client timers. Closing or refreshing the tab within 6 seconds of clicking Delete cancels the pending deletion to prevent accidental data loss.

- **Free-Tier Cold Starts:** Hosting on free-tier platforms (Vercel and Supabase) may exhibit a brief initial wake-up latency after extended inactivity.
- **1,000 Task Display Limit:** For peak browser rendering performance, database queries enforce a 1,000 task ceiling.

5. Important notes or warnings for users

Critical User Warnings:

- **Account Separation:** Signing in with a different Google account opens a completely separate, empty workspace. Your tasks belong exclusively to the specific Google account used during creation.
- **Permanent Deletion:** Once the 6-second undo toast notification expires, deleted tasks are permanently erased from the PostgreSQL database and cannot be recovered.
- **Do Not Store Sensitive Credentials:** Clearpane is intended for everyday productivity. Never store passwords, banking details, private keys, API secrets, or sensitive personally identifiable information (PII) in task titles or notes.
- **Evaluation Build:** Clearpane was developed as an assessment project demonstrating full-stack engineering, performance, security, and accessibility standards.

6. Setup instructions

Follow these instructions to run the complete Clearpane application on your local machine:

System Prerequisites

- Node.js 20+ and npm installed
- Git version control
- A free account at Supabase.com
- A project configured in the Google Cloud Console for OAuth

Step 1: Clone Repository & Install Dependencies

```
git clone https://github.com/Pranesh-DP/kovai.io.git
cd kovai.io
npm install
```

Step 2: Database Schema & Row-Level Security

1. Log in to your Supabase dashboard and open the **SQL Editor**.
2. Paste the contents of `supabase/migrations/0001_tasks.sql` into the editor.
3. Click **Run** to execute the migration:
   - Creates the `tasks` table and `task_status` enum.
   - Installs the `touch_task()` trigger with fixed search path.
   - Enables Row-Level Security (RLS) with SELECT, INSERT, UPDATE, and DELETE policies.

Step 3: Configure Google OAuth 2.0 Credentials

1. In Google Cloud Console $ ightarrow$ **APIs & Services** $ ightarrow$ **Credentials**:
   - Create an **OAuth 2.0 Client ID** (Web Application).
   - **Authorised JavaScript origins:** `http://localhost:3000`
   - **Authorised redirect URIs:** `https://<your-supabase-project-ref>.supabase.co/auth/v1/callback`
2. In Supabase $ ightarrow$ **Authentication** $ ightarrow$ **Providers** $ ightarrow$ **Google**:
   - Enable the Google provider.
   - Paste your Google **Client ID** and **Client Secret**, then click **Save**.
3. In Supabase $ ightarrow$ **Authentication** $ ightarrow$ **URL Configuration**:
   - Set **Site URL** to `http://localhost:3000`.
   - Add `http://localhost:3000/**` to **Redirect URLs**.

Step 4: Configure Local Environment Variables

Create a `.env.local` file in your project root:

```
NEXT_PUBLIC_SUPABASE_URL=https://<your-supabase-project-ref>.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=<your-supabase-anon-or-publishable-key>
NEXT_PUBLIC_SITE_URL=http://localhost:3000
```

Step 5: Run the Development Server

```
npm run dev
```

Open http://localhost:3000 in your browser.

Step 6: Execute Quality Gates & Test Suites

```
# Typecheck
npm run typecheck

# ESLint audit
npm run lint

# Unit & Component Tests (55 tests)
npm run test

# End-to-End Tests (Playwright)
npm run test:e2e
```